

▼ Clustering Analysis

Lecture 5 supplement 2

August 2026

- 01211373 Machine Learning and Programming for Industry

Clustering Analysis

- Unsupervised learning
- Commonly-used algorithms
 - **K-means** (covered in this lecture)
 - **DBSCAN** (covered in this lecture, in separated slides)
 - GMM (Optional : Gaussian Mixture Model)
 - EM (Optional : Expectation-Maximization)

K-means clustering algorithm

The k-means clustering algorithm is as follows:

1. Initialize **cluster centroids** $\mu_1, \mu_2, \dots, \mu_k \in \mathbb{R}^d$ randomly.

2. Repeat until convergence: {

For every i set

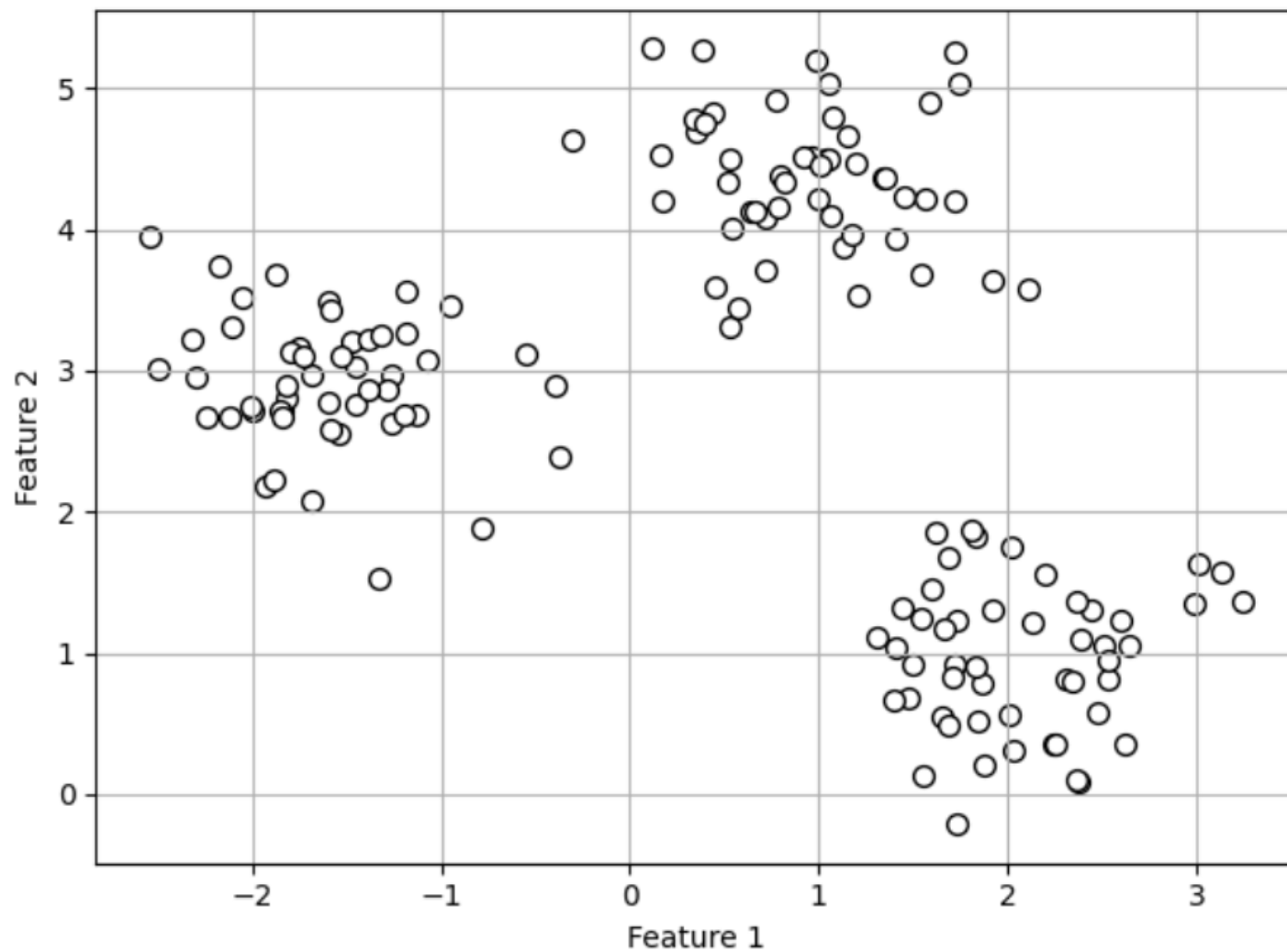
$$c^{(i)} := \arg \min_j \|x^{(i)} - \mu_j\|^2.$$

For each j set

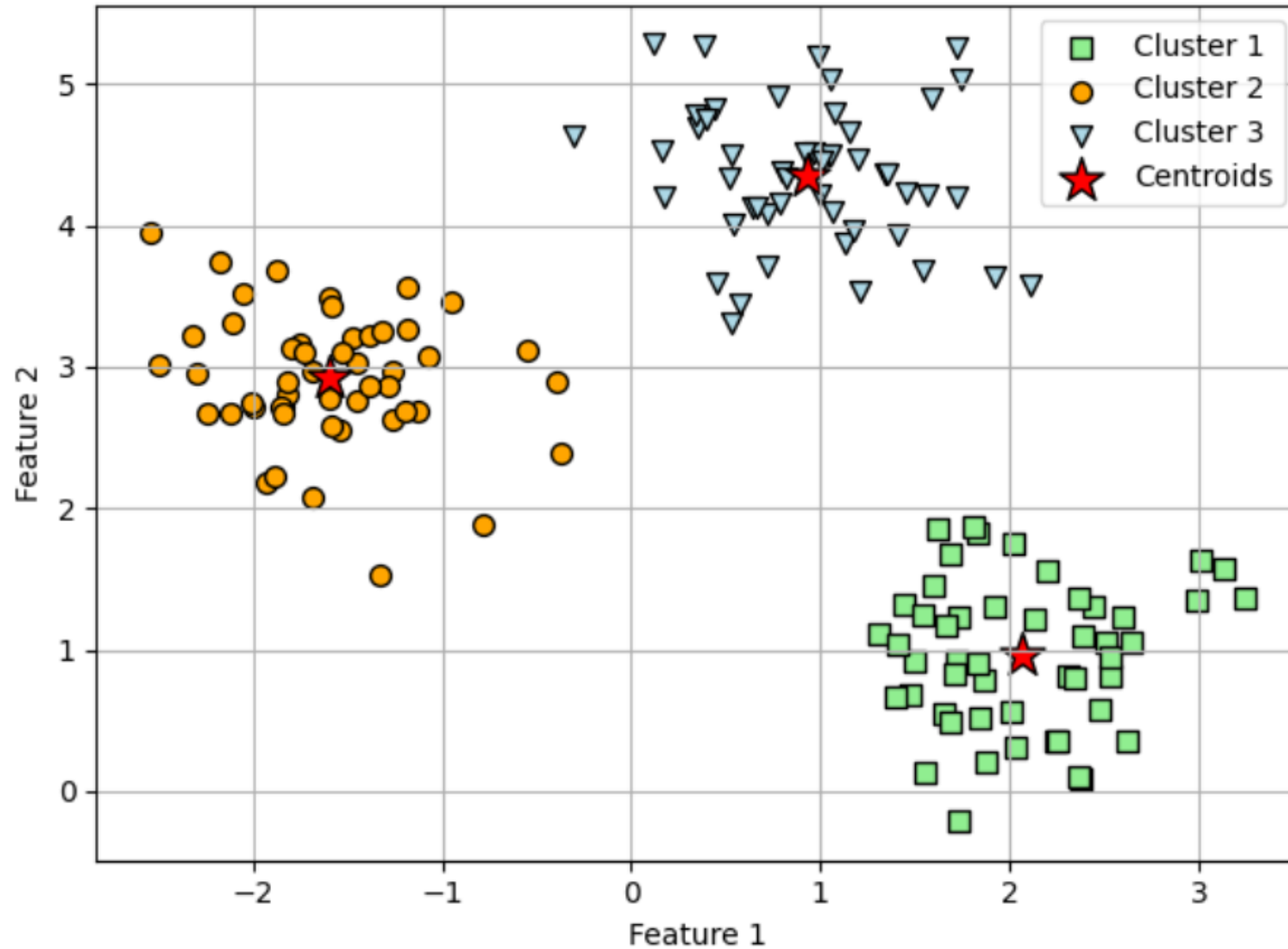
$$\mu_j := \frac{\sum_{i=1}^n 1\{c^{(i)} = j\} x^{(i)}}{\sum_{i=1}^n 1\{c^{(i)} = j\}}$$

}

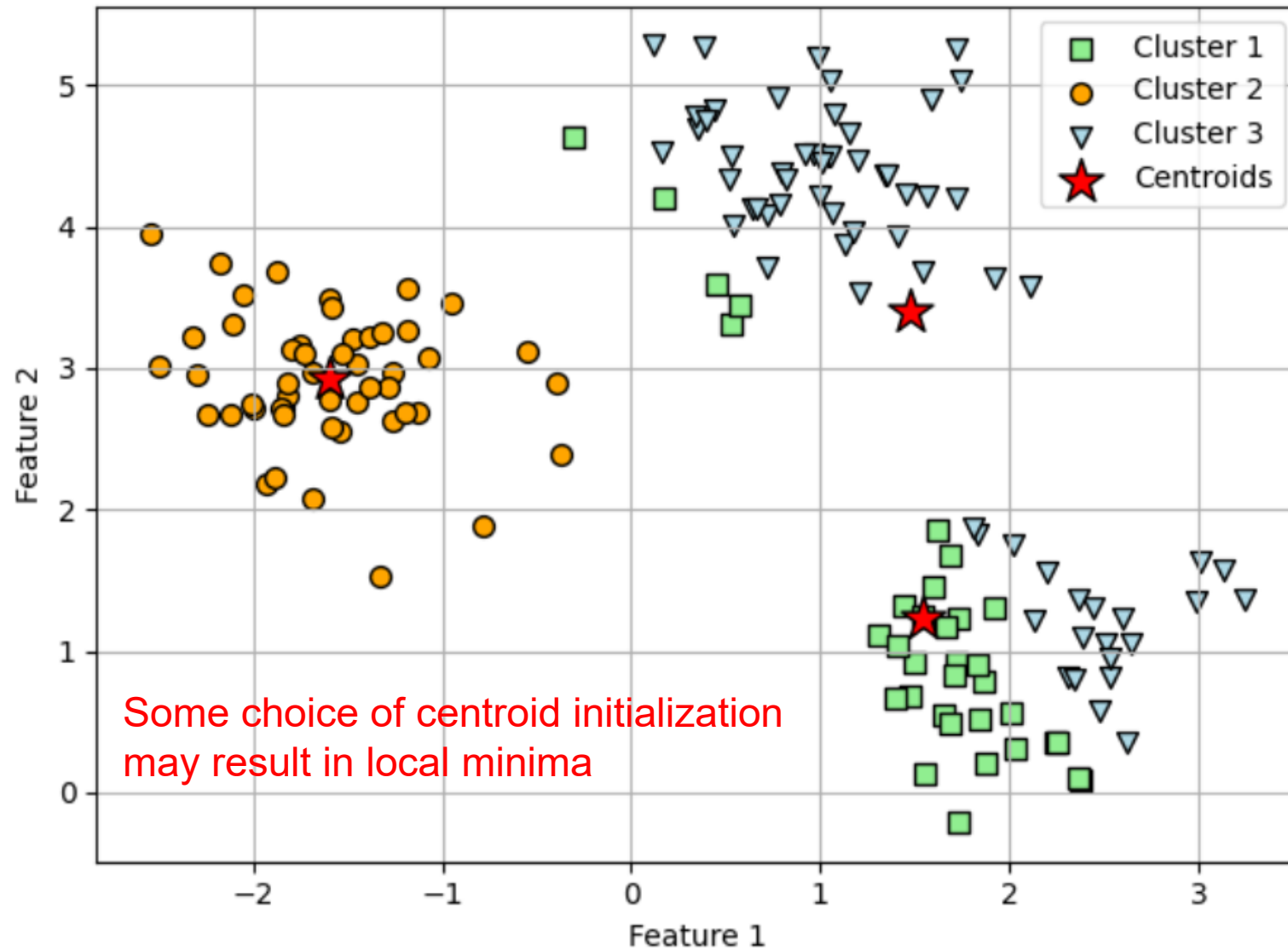
Example : ($k=3$)



Example : (k=3)



Example : ($k=3$)



K-means ++ algorithm (scikit-learn)

- Find some proper way to initialize the centroids by trying to spread them further apart
- Some choices may still result in local minima
- One can try a few choices of initialization and select the one that gives best result

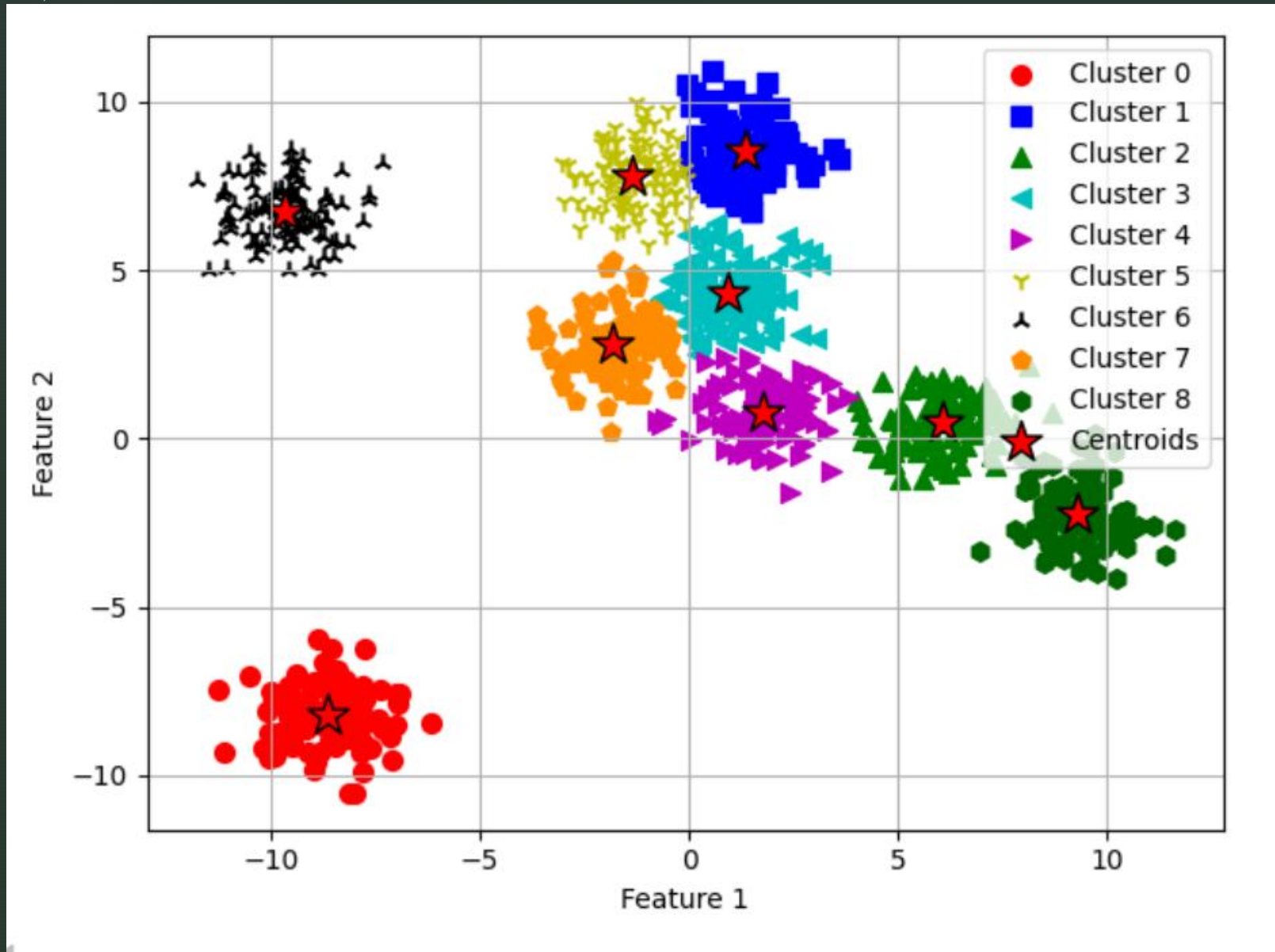
K-means++ algorithm

The initialization in k-means++ can be summarized as follows:

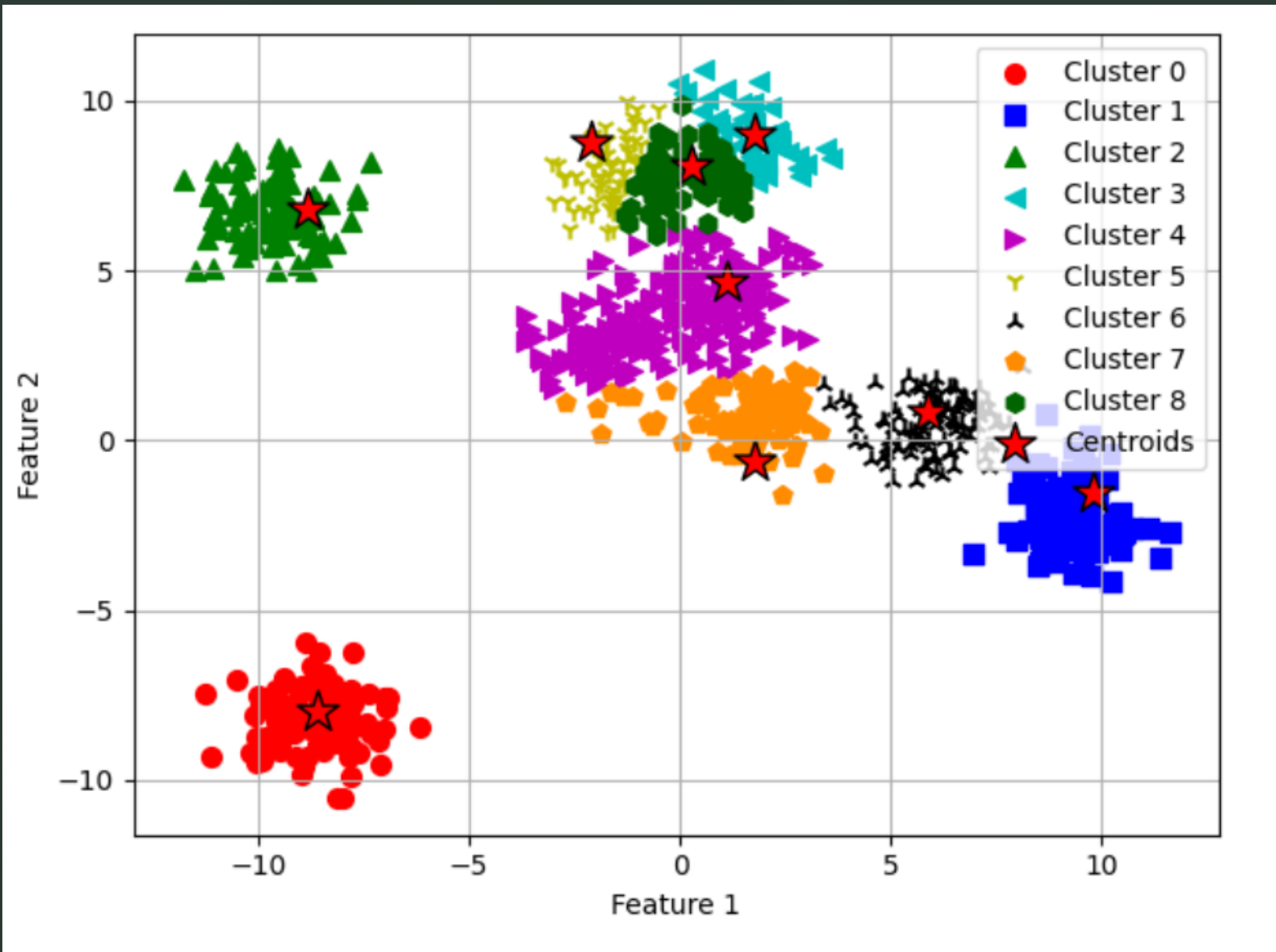
1. Initialize an empty set, $\mathbf{M}$, to store the k centroids being selected.
2. Randomly choose the first centroid, $\mu^{(j)}$, from the input examples and assign it to $\mathbf{M}$.
3. For each example, $x^{(i)}$, that is not in $\mathbf{M}$, find the minimum squared distance, $d(x^{(i)}, \mathbf{M})^2$, to any of the centroids in $\mathbf{M}$.
4. To randomly select the next centroid, $\mu^{(p)}$, use a weighted probability distribution equal to $\frac{d(\mu^{(p)}, \mathbf{M})^2}{\sum_i d(x^{(i)}, \mathbf{M})^2}$. For instance, we collect all points in an array and choose a weighted random sampling, such that the larger the squared distance, the more likely a point gets chosen as the centroid.
5. Repeat *steps 3 and 4* until k centroids are chosen.
6. Proceed with the classic k-means algorithm.

(from page 311 of [1])

Example : (k=9)



Example : (k=9)





Optional materials : P2 of homework 5 (2025) asks you to
implement your own k-means++ algorithm

Hint : you may ask AI to
“implement k means++ algorithm class from scratch python”
and modify the code a little bit to suit the homework

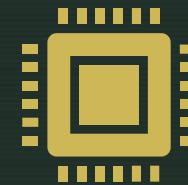
Selected videos



CS229-2022 : Lecture 12 : K-means,
GMM and EM
<https://youtu.be/LMZZPneTcP4?si=yyNtXnx5lwpj-urN>



MIT Machine learning 6.036 : Lecture
13 : Clustering
<https://youtu.be/BaZWcSq3lul?si=jbusFioOt6eU2JBa>



University of Tübingen Machine
Learning I : 09 - Clustering and
expectation-maximization
https://youtu.be/mU3GZaOoVDA?si=3j_tW6OXR6mFg3Me

references

- S. Raschka, Y. Liu and V. Mirjalili. [Machine Learning with PyTorch and Scikit-Learn](#). Packt Publishing. 2022.
- A. Ng. [CS229 Lecture Notes](#). Stanford University. 2023.
- D. Kobak. [Machine Learning I](#). University of Tübingen. 2021.